

Kubernetes - Installation & Setup

Kubernetes: concept, Quality Engineering use cases, first steps and official reference.

Goal of this manual

This manual guides you through the complete installation and configuration process, from preparing the environment to final validation. Follow the steps in order and use each verification checkpoint to confirm that the setup was completed correctly.

Before you start

Confirm the supported operating system, runtime/dependencies and version requirements in the current official documentation before installing. The technical checklist below mirrors the validated tool-specific procedure.

Where to run commands

On Windows use PowerShell or Command Prompt; on macOS/Linux use Terminal. After changing PATH or environment variables, close and reopen the terminal before validating again.

Official references

<https://kubernetes.io/docs/setup/>

Step-by-step installation and setup

1. Follow this tool-specific step: Instale kubectl pela documentação oficial.
2. Follow this tool-specific step: Valide kubectl version --client.
3. Follow this tool-specific step: Prepare um cluster de laboratório pelo método escolhido.
4. Follow this tool-specific step: Execute kubectl config get-contexts e identifique o contexto ativo.
5. Follow this tool-specific step: Execute kubectl cluster-info.
6. Follow this tool-specific step: Execute kubectl get nodes e confirme ao menos um node Ready.
7. Follow this tool-specific step: Faça um deploy simples apenas em ambiente de laboratório e valide com kubectl get pods.

Commands and validation

```
kubectl version --client
```

```
kubectl config get-contexts
```

```
kubectl cluster-info
```

```
kubectl get nodes
```

```
kubectl get pods
```

First working use

Complete the first working use described in the steps above and confirm the expected tool-specific result before adding integrations or advanced configuration.

Common problems and fixes

kubectl funciona mas cluster-info falha: problema é contexto/cluster/credencial, não instalação do CLI.

contexto errado: use kubectl config use-context conscientemente.

x509/Unauthorized: revise kubeconfig, certificados e credenciais do cluster.

Completion criteria

Treat setup as complete when the version/execution checks work and the first use in this manual finishes without errors. Only then move to integrations or advanced configuration.